



Short communication

Assessment of mucosal integrity by quantifying neutrophil granulocyte influx in murine models of acute intestinal injury

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ABSTRACT

Intact epithelial body surfaces represent physical barriers which protect the organism from invading pathogens and loss of nutrients. Barrier malfunction is closely linked to disorders such as inflammatory bowel disease and graft-versus-host disease. In fact, several pharmacological or radiobiological therapeutic strategies have side effects that affect epithelial surfaces. In this context, assays that accurately assess epithelial barrier integrity in patients and animal models are crucial to create a better understanding of the mechanisms leading to disease or limiting therapeutic approaches due to barrier disruption. Here, we tested the ability of the widely used FITC-dextran intestinal permeability analysis to evaluate loss of intestinal barrier integrity in different murine models of gut mucosal damage and established influx of neutrophil granulocytes into the intestinal lamina propria (LP) as an alternative approach. We demonstrate that the sensitivity and specificity of FITC-dextran intestinal permeability analysis is relatively low: Although it did represent severe forms of mucosal damage due to intensive conditioning therapy (high doses of either total body irradiation (TBI) or chemotherapy) or after conditioning and allogeneic stem cell transplantation, it did not recognize less severe forms of damage as after lower doses of TBI or chemotherapy alone. In addition, discrimination of untreated from irradiated mice by differences in FITC-dextran translocation was not exact. In contrast, influx of neutrophil granulocytes into the intestinal LP, which reflects immune activation due to translocation of microbes and microbial products during intestinal barrier breach, quantitatively correlated with the severity of intestinal barrier damage. It accurately represented both severe and less severe forms of intestinal damage as after high or lower dose TBI or chemotherapy and correctly discriminated treated from untreated animals. Taken together, we demonstrate the limitations of FITC-dextran intestinal permeability analysis and identify intestinal neutrophil influx as a powerful additional tool to measure breakdown of intestinal barrier function.

1. Introduction

Mucosal barriers like the intestinal epithelial cell layer protect sterile compartments from physical, chemical and microbial challenge [1]. Inherited or acquired barrier malfunction is closely linked to disorders like inflammatory bowel disease (IBD) [1,2]. In addition to genetically driven diseases and barrier dysfunction resulting from chronically disrupted mucosal homeostasis, acute exogenous triggers can also cause damage to the intestinal barrier and favor the pathogenesis of illnesses like IBD or Graft-versus-host disease (GVHD), which can follow allogeneic stem cell transplantation (allo-HSCT) [2,3]. GVHD occurs in as many as 65% of transplant recipients and is caused by alloreactive T cells that attack host tissue, predominantly of the gut, liver and skin [4]. A crucial step in the development of acute GVHD is pre-transplant conditioning therapy by chemotherapy (CTX) or total

body irradiation (TBI). Both can result in acute breakdown of intestinal barrier function followed by translocation of microbes and Pathogen-associated molecular patterns (PAMPs) into underlying sterile compartments. Activated intestinal immune cells prime donor T cells which attack host tissue and augment intestinal damage [4,5]. Although such collateral damage to mucosal integrity limits several pharmacological or radiobiological therapeutic strategies, recent evidence shows that certain aspects of collateral intestinal barrier disruption can be beneficial as there seems to be a complex interplay between side effects and direct anti-tumor effects of chemotherapeutic tumor therapy: In certain settings, efficient tumor control requires intestinal microbiota and reduced intestinal integrity during chemotherapy [6]. To create a better understanding of the mechanisms leading to enhanced or reduced intestinal integrity, it is crucial to be able to quantify changes in barrier function sufficiently. To date, efforts to evaluate the intestinal

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barrier function in patients mainly rely on biomarkers (e.g. fecal calprotectin, fecal albumin), biopsies (e.g. tight junction protein expression) and diagnostic permeability tests (lactulose/mannitol urinary excretion etc.), all of which are flawed by certain limitations [2,7]. Several assays are used to measure intestinal permeability in mice [7]. FITC-dextran-intestinal permeability analysis, for example, uses dextran molecules as inert, non-digestible test probes of variable size linked to the fluorescent marker fluorescein isothiocyanate (FITC). In mice with a severely damaged intestinal mucosa, FITC-dextran molecules that were applied orally can leave the gut lumen and enter the systemic blood circulation, where FITC-fluorescence can be measured. This assay is used to assess intestinal permeability during gastrointestinal GVHD after allo-HSCT [8] as well as in models of colitis, hypoxia, liver disease, trauma and CTX [6,7]. Nevertheless, clear descriptions of the characteristics and limitations of FITC-dextran intestinal permeability analysis are lacking. In this report, we discuss our experience with the FITC-dextran permeability assay to measure intestinal integrity after TBI, CTX and allo-HSCT. We demonstrate that this assay suffers from limited sensitivity and specificity to detect loss of intestinal barrier integrity as after reduced doses of TBI. Alternatively, we propose quantitative evaluation of neutrophil influx into the intestinal lamina propria (LP) as a powerful additional tool to more accurately evaluate intestinal integrity. We confirm that neutrophil chemotaxis depends on intestinal microbiota [9] and show that neutrophil influx quantitatively correlates with the severity of acute intestinal tissue damage, offering higher sensitivity and specificity when using FITC-dextran intestinal permeability analysis.

2. Methods

2.1. Mice

C57BL/6 and Balb/c mice were purchased from Janvier Labs (France). Mice were used between 6 and 8 weeks of age (onset of experiments) and were maintained in specific pathogen free conditions. Animal studies were approved by the local regulatory agencies (Regierung von Oberbayern, Munich).

2.2. Models of intestinal mucosal damage

2.2.1. Intestinal damage by total body irradiation + allo-HSCT

Allo-HSCT was performed as previously described [10]. Briefly, Balb/c recipient mice received 2×4.5 Gy TBI and were then given 5×10^6 T cell-depleted donor (C57BL/6) bone marrow (TCD-BM) cells $\pm 0.5 \times 10^6$ donor T cells to induce acute GVHD (CD4/CD8 enrichment, Miltenyi). T cell depletion of BM cells was performed using CD90.2 MACS sorting beads (Miltenyi). Total TBI and T cell doses were chosen to ensure a strong T cell mediated damage within one week after transplantation [11]. TBI was split into two doses (2×4.5 Gy) to reconcile reduced collateral toxicity with optimal eradication of recipient/engraftment of donor hematopoietic compartments [12].

2.2.2. Intestinal damage by total body irradiation alone

6–8 weeks old Balb/c or C57BL/6 mice were irradiated with different TBI doses (0–11 Gy) to achieve different levels of damage to the intestinal mucosa, as indicated in the text and figure legends. The maximal TBI doses (9 Gy Balb/c, 11 Gy C57BL/6) were chosen according to commonly applied myeloablative TBI doses (as used for conditioning therapy before allo-HSCT of respective recipients) depending on the radio sensitivity of the mouse strain used [11].

2.2.3. Intestinal damage by chemotherapy alone

C57BL/6 mice were treated with Doxorubicin injected intraperitoneally (7.5 or 20.0 mg/KG body weight) as indicated in the text and figure legends.

2.3. Intestinal FITC-dextran intestinal permeability analysis

FITC-dextran intestinal permeability analysis was performed similar to what was previously described [8]. Mice were kept without food and water for 8 h. Then, FITC-dextran (#FD4-1G, Sigma) was administered by oral gavage at a concentration of 50 mg/ml in water (750 mg/kg). 4.5 h later, plasma was collected from peripheral blood (8800 rcf, 10 min), then mixed 1:1 with PBS and analyzed on a plate reader at 485 nm excitation wavelength and 535 nm emission wavelength.

2.4. Assessment of intestinal neutrophil granulocyte influx

Mice were sacrificed, lamina propria leukocytes (LPL) were isolated and neutrophils within LPL were analyzed by flow cytometry. Isolation of lamina propria leukocytes was performed as previously described [10]. Colon and/or ileum (defined as distal 1/3 of small intestine) were flushed with cold PBS and cut into 2 cm pieces. Longitudinally opened intestines were washed and then incubated with HBSS solution containing 2 mM EDTA, 10 mM HEPES, 5% FCS (Hyclone), 1% Penicillin-Streptomycin, 1% L-Glutamine and 1 mM DTT (all Sigma-Aldrich). After incubation on a shaker (225 rpm) at 37 °C for 15 min, tissues were washed and filtered through a 100 μ m strainer (BD 352360). Next, intestines were incubated for 45 min in PBS⁺Ca⁺⁺Mg supplemented with FCS (10%), Collagenase II (200 U/ml; Worthington), and DNase I (0.05 mg/ml; Roche) on a shaker at 37 °C. Cells in suspension were filtered through a 100 μ m strainer and purified on a 40/80% Percoll gradient (Biochrom).

2.5. Flow cytometry

Cell suspensions were stained in PBS with 3% FCS. After live/dead staining (Cat-No: 65-0866-18, eF506, ebioscience) and Fc-Block (Cat-No 101330, BioLegend) according to manufacturer's instructions, indicated clones of fluorochrome-coupled anti-mouse antibodies (BioLegend) were used for surface staining: CD11b (M1/70, APC), CD11c (N418, FITC), Gr-1 (RB6-8C5, PerCP-Cy5.5), Ly6G (1A8, PE-Cy7), CD19 (1D3/CD19, PE), CD3 (17A2, PE), Ly6C (HK1.4, APC-Cy7), NK1.1 (PK136, PE). Neutrophils were generally identified as CD11b^{hi} Gr-1^{hi} live cells (simplified analysis). In the experiments depicted in [Supplementary Fig. 2](#), an extended analysis using CD11b^{hi} Ly6G⁺ CD3[−] CD11c^{hi} CD19[−] Ly6C^{int} NK1.1[−] live cells [13] was performed to validate simplified analysis for the general purpose of this manuscript. Stainings performed are indicated in the figure legends. Data was acquired on a FACS Canto II (BD Biosciences) and analyzed using FlowJo software (TreeStar).

2.6. Reduction of intestinal bacterial load

Reduction of intestinal bacterial load was performed with C57BL/6 mice as previously described [14]. 6 weeks old C57BL/6 mice were given a combination of four antibiotics in the drinking water for at least 4 weeks: vancomycin (0.5 g/L), ampicillin (1.0 g/L), neomycin (1.0 g/L) and metronidazole (1.0 g/L). Reduction of intestinal bacterial load was confirmed after incubation of supernatants of diluted fecal pellets on Schaedler Agar (aerobic incubation) and Columbia Agar (anaerobic incubation).

2.7. Statistics

GraphPad Prism version 6 was used for statistical analysis. Statistical tests applied to evaluate differences between means of experimental groups are indicated in the respective figure legends. Experiments were analyzed using two-tailed unpaired *t* test or ordinary one-way Anova, corresponding to the distribution shape of our observations. We used ordinary one-way Anova for multiple comparisons and performed Dunnett's test for Multiple-test corrections. P

values less than 0.05 were considered to be statistically significant.

For further statistical analysis it was assumed that observations $X_1, \dots, X_n$ from untreated and treated mice were i.i.d. (independent and identically distributed). To be more specific, each random sample was considered to be normally distributed with unknown mean μ and variance σ^2 . The estimation of these parameters was performed by applying standard maximum-likelihood techniques, i.e., the probability to exactly obtain the given sample was maximized. This two-dimensional maximization problem yields the standard estimators $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i$ and $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \hat{\mu})^2$. In a second step, these estimates were used to obtain acceptance regions in statistical tests to allow for distinguishing healthy untreated from treated mice. With the null hypothesis H_0 , the observation originates from an untreated mouse, the common test errors are defined as follows: A type-I-error (α -error) occurs when erroneously identifying an untreated mouse as treated and a type-II-error (β -error) falsely identifies a treated mouse as untreated. Setting the type-I-error to a specific level α directly yields a cut-off value for accepting or rejecting the null hypothesis. Here, this is simply the $(1-\alpha)$ -quantile of the respective Gaussian distribution under the null hypothesis. Once the cut-off value was determined, both the type-II-error and the power of the test $1-\beta$ could be obtained. Note that the power of a statistical test is the key property to compare the significance of different tests with same level α . In the sequel, we also refer to the terms *expected* sensitivity ($1-\beta$) and specificity ($1-\alpha$) to emphasize that in future test settings it might not be known if mice are treated or not. Note that commonly the terms sensitivity and specificity are applied in the context of relative frequencies, thus being able to verify the null hypothesis with independent diagnostics.

3. Results

3.1. Translocation of FITC-dextran detects barrier loss with low sensitivity and specificity

First, we assessed the ability of a commonly used tool to evaluate the integrity of mucosal barrier function, the FITC-dextran permeability analysis. To do so, we measured FITC-dextran fluorescence in the blood following oral application in a mouse model of allo-HSCT after conditioning with TBI. Allo-HSCT was performed with Balb/c recipients after split dose TBI (2×4.5 Gy), which is optimal to eradicate recipient and allow engraftment of donor hematopoietic compartments with reduced collateral damage. As expected [8,15], we observed a robust increase of FITC-dextran serum levels in recipient mice 7 days after allo-HSCT with donor T cell-depleted BM + T cells (TBI 2×4.5 Gy + TCD-BM + T cells). In contrast, there was no significant difference in translocated FITC-dextran levels on day +7 between healthy untreated mice (no allo-HSCT) and mice that had received TBI and donor BM but no donor T cells (TBI 2×4.5 Gy + TCD-BM control mice) (Fig. 1A). These data suggest that FITC-dextran intestinal permeability analysis detects severe forms of mucosal damage (split dose TBI + allo-HSCT with T cells) but is less sensitive for less severe forms of mucosal damage (split dose TBI only). To address whether FITC-dextran intestinal permeability analysis can detect radiation-induced intestinal barrier damage at all, we next assessed FITC-dextran translocation after single dose 9 Gy TBI, which is expected to be more disruptive than split dose TBI [12]. Furthermore, we analyzed FITC-dextran levels in the serum on day + 3, which is when we expect maximum levels after TBI. Although irradiated mice showed significantly increased FITC-dextran serum levels (Fig. 1B) 3 days after 9 Gy TBI, those levels were much lower than 7 days after TBI 2×4.5 Gy + TCD-BM + T cells. These results were in line with the current understanding that TBI leads to intestinal barrier dysfunction and translocation of bacteria and PAMPs, followed by an inflammatory response which augments intestinal tissue damage through activated allogenic T cells [4,5]. To statistically evaluate the estimated sensitivity and specificity of FITC-dextran

intestinal permeability analysis that can be expected in further experiments, we fitted the probability density functions (Fig. 1C) of FITC-dextran serum levels after 9 Gy TBI vs no treatment from Fig. 1B, using maximum-likelihood estimates (see Section 2.7) and assuming that values in both groups were normally distributed. We illustrated our results in an ROC curve, a common way to illustrate the power of a test for different levels α (for details see Section 2.7), and studied the statistical inference of using various cut-off values (Supplementary Fig. 1A). Setting the level α to a usual threshold of 5% led to a cut-off value of 2505 ng/ml, resulting in a type-II-error of 23% (false negatives) and related power of 77%. While 5% of the untreated mice would be misidentified as irradiated mice (false positives), 77% of all irradiated mice would be correctly identified (Fig. 1C). Such rather low expected values of sensitivity (77%) and specificity (95%) (Table 1) indicated that FITC-dextran translocation may not represent intestinal barrier dysfunction with sufficient accuracy in this setting.

3.2. Neutrophil granulocyte migration into the intestinal lamina propria reflects barrier breach and translocation of microbiota

A recent publication demonstrated that TBI leads to neutrophil granulocyte chemotaxis into the intestinal LP. Since neutrophil influx was dramatically reduced in germfree mice [9], it was hypothesized that reduced intestinal integrity after TBI leads to translocation of bacteria and PAMPs which results in an inflammatory and chemotactic response with recruitment of neutrophils into the LP. We could confirm that relative amounts of neutrophils (as percentage of total lamina propria leukocytes) in the small and large intestine are strongly increased after TBI and observed the peak of this dynamic process on day three after TBI (Fig. 2A). We observed that about 97% of infiltrating CD11b^{hi} Gr-1^{hi} cells are in fact Ly6C^{int} Ly6G⁺ neutrophil granulocytes but not Ly6C^{hi} Ly6G^{lo} macrophages or other immune cells (Supplementary Fig. 2A, B, C) [13]. Furthermore, we corroborated the assertion that translocation of bacteria and PAMPs contribute to this process since mice that had received antibiotics in the drinking water to reduce the intestinal microbial burden showed significantly decreased neutrophil levels after TBI compared to mice that had not received antibiotics (Fig. 2B).

3.3. Lamina propria neutrophil analysis is sensitive and specific

Next, we examined the power of lamina propria neutrophil analysis as a read-out to discriminate between untreated and irradiated mice (Fig. 2C). Here as well, the percentages of neutrophils of all LPL of untreated and irradiated mice were assumed to be Gaussian. The probability density functions of lamina propria neutrophil levels of healthy untreated and irradiated mice were fitted as described before (Fig. 2D). Accepting a very low type-I-error of 0.001% (false positives) yields a cut-off value of 0.45% CD11b^{hi} Gr-1^{hi} and a very high power of 99.995% (Fig. 2D, Supplementary Fig. 1B). Thus, nearly all mice were correctly identified as either untreated or treated, indicating the considerable expected sensitivity (99.5%) and specificity (99.9%) (Table 1).

3.4. Lamina propria neutrophil analysis provides an enhanced detection range

We further evaluated the detection range of LP neutrophil vs FITC-dextran intestinal permeability analysis by testing lower dose levels of TBI. In fact, mice that received 5.5 Gy TBI already showed significantly enhanced neutrophil levels compared to untreated mice. Neutrophil levels of mice that received 9 Gy TBI were further significantly increased (Fig. 2E). In contrast, FITC-dextran intestinal permeability analysis did not detect intestinal barrier damage after 5.5 Gy TBI, as FITC-dextran levels in the serum after irradiation with 5.5 Gy were not increased compared to untreated control mice (Fig. 2E). As tight

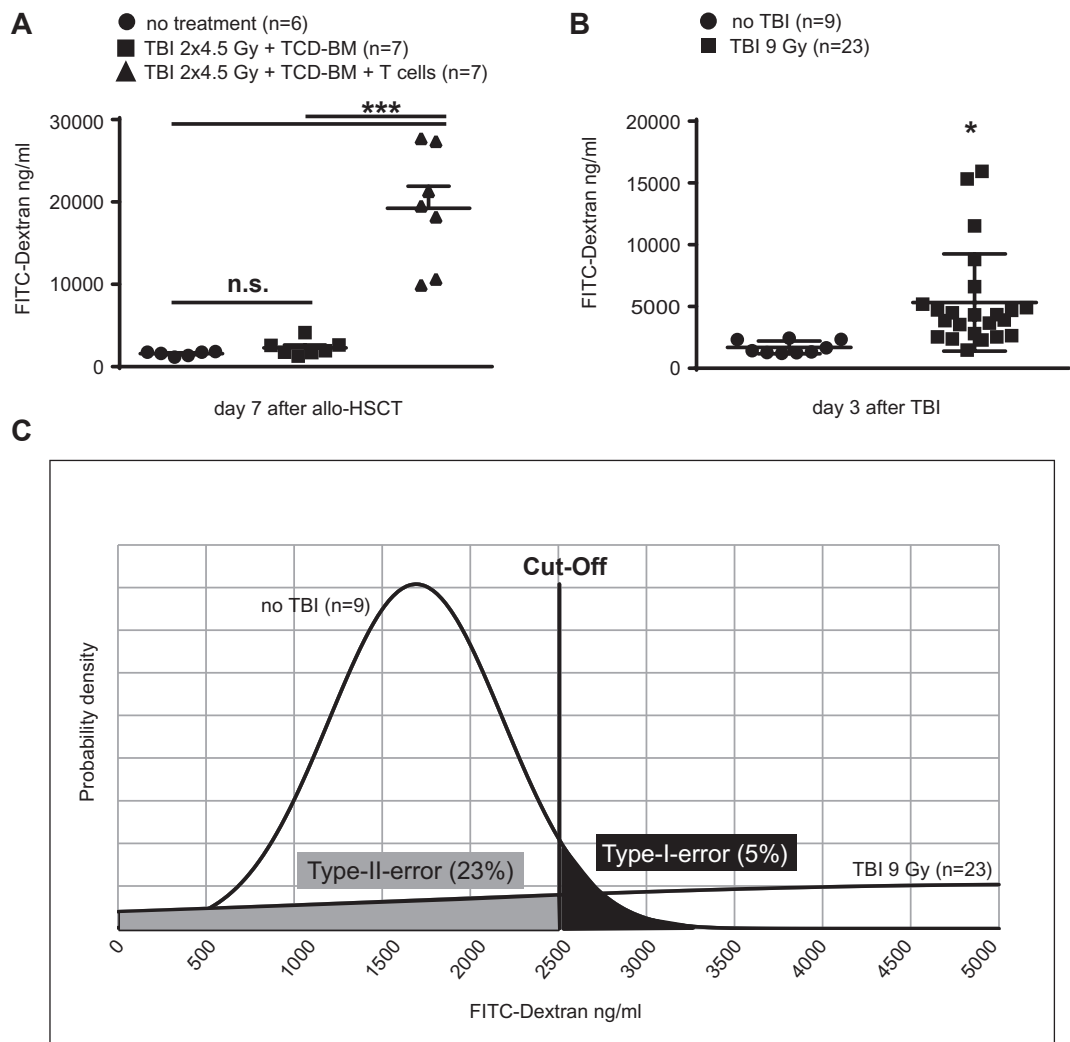


Fig. 1. FITC-dextran intestinal permeability analysis detects barrier loss with low sensitivity and specificity (A) Balb/c mice were either left untreated (n = 6) or received split dose 2 × 4.5 Gy TBI followed by allo-HSCT with T cell-depleted bone marrow (TCD-BM) only (n = 7) or with 2 × 4.5 Gy TBI + TCD-BM + T cells (n = 7) on day 0 and were then challenged with oral gavage of FITC-dextran on day + 7. Depicted are FITC-dextran concentrations in the serum. Data from one representative experiment. (B) Balb/c mice were either left untreated (n = 9) or received single dose 9 Gy TBI (n = 23) on day 0 and were then challenged with oral gavage of FITC-dextran on day + 3. Depicted are FITC-dextran concentrations in the serum. Pooled data of 3 independent experiments. (C) Fitted Gaussian distribution of mice depicted in (B) and statistical analysis of Type-I-error (5%) and resulting Type-II-error. Experiments were analyzed using parametric two-tailed unpaired t test and ordinary one-way Anova for multiple comparisons. Significance was set at P values < 0.05, p < 0.01 and p < 0.001 and was then indicated with asterisks (*, ** and ***). Data are presented as mean ± S.E.M.

Table 1
Detection of mucosal damage by high dose TBI.

Test characteristics:	Sensitivity	Specificity
FITC dextran intestinal permeability analysis	77%	95%
Neutrophil infiltration analysis	99.5%	99.9%

junctions are central constituents of epithelial barrier function in the gut, we next analyzed the effect of different dose levels of TBI on the gene expression levels of several relevant tight junctional proteins [2,16,17]. We observed several significant changes in gene expression levels of Claudin 2, Claudin 4, Claudin 11, Occludin, JAM-A and ZO-1 between untreated mice and mice that received 5.5 Gy or 9 Gy TB, respectively (Supplementary Fig. 3). Irradiation is known to affect mRNA expression levels of tight junctional proteins differentially with respect to time after exposure and intestinal segment affected [16]. Whether these changes in mRNA expression of tight-junctional proteins translate into differential protein expression and whether they reflect damage or regeneration of barrier function remains to be determined. We believe that dose-sensitive, differential mRNA expression of tight-

junctional proteins reflects individual damage levels to the intestinal barrier integrity which can be assessed by LP neutrophil quantification.

3.5. Lamina propria neutrophil analysis is versatile

To establish that LP neutrophil quantification can be used to measure different levels of damage to intestinal barrier integrity in general, we used an alternative damage model with CTX. First, we analyzed FITC-dextran translocation into the blood after doxorubicin treatment i.p. to show that intestinal integrity was affected. High dose doxorubicin (20 mg/kg body weight) led to significantly increased FITC-dextran levels in the serum, whereas reduced dose doxorubicin (7.5 mg/kg) did not (Fig. 3A). In contrast, animals that received doxorubicin showed significantly enhanced influx of neutrophils into the small intestinal LP in both treatment groups. As with TBI, pretreatment with oral antibiotics also reduced neutrophil levels after doxorubicin treatment (Fig. 3B). Quantitative representation of different levels of intestinal damage was thus similar after CTX and after TBI (Figs. 3A and 2E), indicating the versatility of this test.

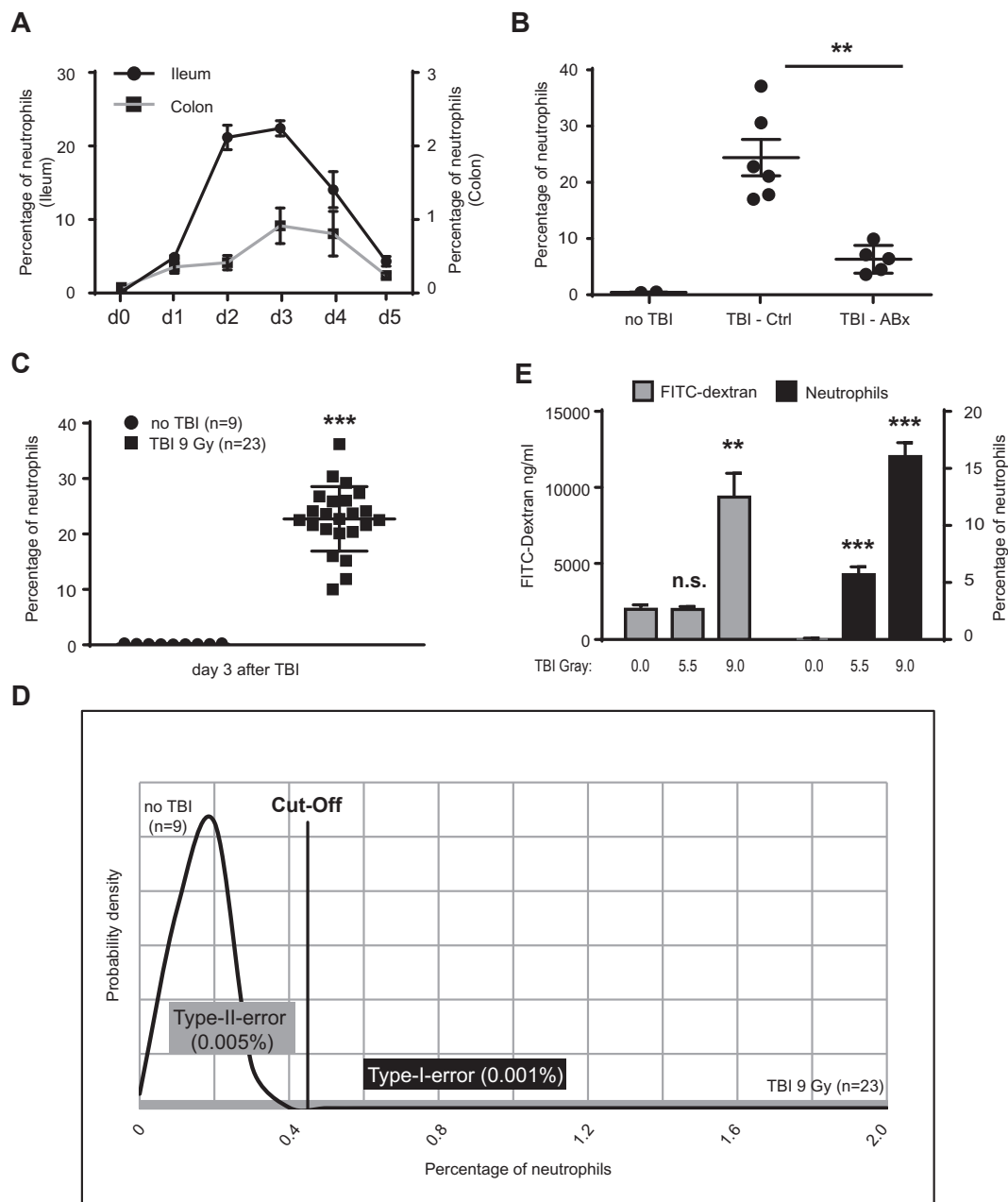


Fig. 2. Quantification of neutrophil granulocytes in the intestinal lamina propria detects barrier breach with high sensitivity and specificity. (A) Lamina propria leukocytes (LPL) isolated from small (Ileum) and large intestines of Balb/c mice were analyzed by flow cytometry at different time points after 9 Gy TBI. Percentages of neutrophils (CD11b^{hi} Gr-1^{hi}) of all live cells are depicted. D0 (n = 2), d1 (n = 3), d2 (n = 3), d3 (n = 4), d4 (n = 4) and d5 (n = 4). Pooled data of 2 independent experiments. (B) C57BL/6 mice were treated with antibiotics for 4 weeks (n = 5) or left untreated (n = 6) and then received 11 Gy TBI. On day 3 after TBI LPL isolated from small intestines (Ileum) were analyzed by flow cytometry. Percentages of neutrophils (CD11b^{hi} Gr-1^{hi}) of all live cells are depicted. Pooled data of 2 independent experiments. (C) LPL isolated from small intestines (Ileum) of Balb/c mice were analyzed by flow cytometry on day 3 after 9 Gy TBI (n = 23) and were compared with untreated mice (n = 9). Percentages of neutrophils (CD11b^{hi} Gr-1^{hi}) of all live cells are depicted. Pooled data of 3 independent experiments. (D) Fitted Gaussian distribution of mice depicted in (C) and statistical analysis of Type-I-error (0.001%) and resulting Type-II-error (0.005%). (E) Untreated Balb/c mice (n = 8), mice that received 5.5 Gy (n = 8) or 9 Gy (n = 8) TBI were challenged with oral gavage of FITC-dextran on day 3. FITC-dextran concentrations in the serum and percentages of neutrophils (CD11b^{hi} Gr-1^{hi}) of all live LPL isolated from small intestine (Ileum) are depicted. Pooled data of 4 independent experiments. Experiments were analyzed using parametric two-tailed unpaired *t* test and ordinary one-way Anova for multiple comparisons. Significance was set at *P* values < 0.05, *p* < 0.01 and *p* < 0.001 and was then indicated with asterisks (*, ** and ***). Data are presented as mean ± S.E.M.

4. Discussion

In this report, we characterize and compare two techniques to detect loss of intestinal barrier integrity after pharmacological, radiation-induced or immune mediated epithelial injury through CTX, TBI or TBI + allo-HSCT, respectively. FITC-dextran intestinal permeability analysis, commonly used by researchers to evaluate mucosal barrier function, can detect barrier disruption after high doses of CTX or TBI and after TBI + allo-HSCT. However, FITC-dextran levels in the serum

cannot reflect more subtle forms of epithelial damage such as after lower doses of CTX or TBI alone, indicating low sensitivity. Statistical analysis of our observations revealed that even with high doses of TBI, as applied before murine allo-HSCT, this assay does not accurately discriminate between untreated vs irradiated mice. Setting the level α to 5% resulted in a type-II-error (false negatives) of 23%. While 5% of all untreated mice would be misidentified as irradiated mice (false positives), only 77% of all irradiated mice would be correctly identified. Naturally, a test with such low expected sensitivity (77%) and

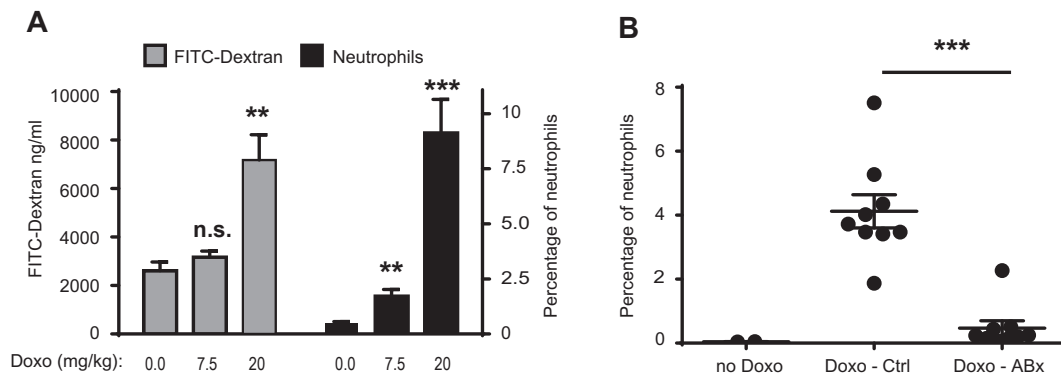


Fig. 3. Quantification of neutrophil granulocytes in the intestinal lamina propria reflects barrier breach after different doses of chemotherapy. (A) Untreated Balb/c mice and Balb/c mice that received Doxorubicin 7.5 mg/kg or Doxorubicin 20 mg/kg i.p. on day 0 were challenged with oral gavage of FITC-dextran on day 3. FITC-dextran concentrations in the serum and percentages of neutrophils (CD11b^{hi} Gr-1^{hi}) of all live LPL isolated from small intestines (Ileum) are depicted. Pooled data of 4 independent experiments with 4–12 mice per group. (B) C57BL/6 mice were treated with antibiotics (n = 9) for 4 weeks or were left untreated (n = 9) and then received Doxorubicin (20 mg/kg) i.p. On day 3 after treatment LPL isolated from small intestines (Ileum) were analyzed by flow cytometry. Percentages of neutrophils (CD11b^{hi} Gr-1^{hi}) of all live cells are depicted. Pooled data of 2 independent experiments. Experiments were analyzed using parametric two-tailed unpaired t test and ordinary one-way Anova for multiple comparisons. Significance was set at P values < 0.05, p < 0.01 and p < 0.001 and was then indicated with asterisks (*, ** and ***). Data are presented as mean ± S.E.M.

specificity (95%) values will be insufficient to detect minor differences in severity of intestinal barrier damage such as it might become relevant when analyzing effects of different modalities and dose levels between clinically relevant experimental procedures and groups. Searching for more reliable parameters, we here established neutrophil influx into the intestinal LP as a sensitive marker, which indeed quantitatively reflected different levels of damage to the mucosal barrier. It detected lower levels of damage and allowed to discriminate different damage intensities as after TBI or CTX, displaying extremely limited expected error rates. Nearly all untreated vs irradiated mice would be identified correctly using this approach. Consequently, expected values for sensitivity (99.5%) and specificity (99.9%) were high. We thus postulate that quantifying neutrophil influx into the intestinal LP is a conclusive and versatile test to detect and represent different levels of damage to the mucosal barrier as might be encountered between (I) transgenic mice with defects that result in enhanced or reduced capacity to maintain intestinal integrity (II) different treatment modalities or toxic insults (III) scenarios that influence the intestinal barrier indirectly via systemic mediators as during shock or sepsis. Interestingly, a well-established biomarker for intestinal inflammation in human IBD is fecal calprotectin, an anti-microbial peptide shed into the intestinal lumen by neutrophil granulocytes that infiltrate the intestinal mucosa. This underscores that quantification of neutrophil influx into the gut LP in mice may be a valid test for intestinal integrity with direct correlations to the human system. A limitation to this approach however is that mucosal neutrophil influx reflects translocation of bacteria or PAMPs across the mucosal barrier and is thus biased by both intestinal microbiota and the functional status of the immune system. Direct measurement of translocated microbes (e.g. by culturing or ribosomal RNA sequencing) or microbial products (e.g. LPS) in the portal or systemic circulation would be an alternative but can be technically challenging, suffer from rather insensitive detection ranges and limited culture options and are prone to contamination.

5. Conclusion

As previously reported, all established tests to evaluate intestinal barrier function have relevant advantages and limitations [2,7]. There is no universal marker that reliably reflects mucosal permeability, a prerequisite for a better understanding of the pathophysiology of systemic inflammatory disorders like IBD or GVHD as well as certain aspects of immune and cancer therapy. A combination of methods might therefore be most useful to assess intestinal barrier integrity [7]. Here we show that the analysis of neutrophil influx into the gut lamina

propria is a sensitive, specific and versatile method to accurately detect and quantify injury to the integrity of the intestinal barrier, appropriate to complement other, less powerful tools such as FITC-dextran intestinal permeability analysis.

Author contributions

J.C.F. designed, performed and analyzed experiments. A.W. helped to perform experiments. T.H. and H.P. guided the study. J.C.F., T.H. and H.P. wrote the manuscript. This work is part of the thesis of J.C.F. at the Technical University of Munich (TUM).

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.cellimm.2017.04.003>.

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